

FILLMORE COUNTY AP, MN, USA

WMO: 720283

Lat: 43.677N

Lon: 92.180W

Elev: 389

StdP: 96.74

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04927

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.7	-21.3	-27.6	0.3	-23.3	-25.3	0.4	-21.1	12.8	-6.4	11.6	-8.5	3.9	310	0.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	30.8	23.0	29.1	21.8	27.6	20.9	24.5	28.5	23.4	27.5	22.4	26.3	5.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	18.8	26.9	22.2	17.6	26.2	21.1	16.5	25.2	75.9	28.5	71.8	27.8	67.8	26.3	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.4	8.5	DB	-27.7	33.3	2.8	1.8	-29.7	34.6	-31.3	35.6	-32.9	36.6	-35.0	37.9	(4)
(5)			WB	-27.9	26.2	2.8	1.4	-29.9	27.2	-31.5	28.0	-33.1	28.8	-35.1	29.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	-8.6	-7.4	0.5	7.8	14.4	19.7	21.6	20.1	16.8	9.4	2.1	-5.6	(6)	
	DBStd	11.94	7.11	6.67	6.78	5.56	4.94	3.22	2.95	3.08	4.23	5.19	5.92	6.21	(7)	
	HDD10.0	2293	578	486	304	106	16	0	0	0	3	74	244	482	(8)	
	HDD18.3	4198	836	719	553	317	141	22	7	17	78	280	488	741	(9)	
	CDD10.0	1433	0	0	9	39	153	290	359	312	208	57	7	0	(10)	
	CDD18.3	296	0	0	1	1	19	62	107	70	32	4	0	0	(11)	
	CDH23.3	2179	0	0	2	16	170	448	813	468	227	35	0	0	(12)	
	CDH26.7	521	0	0	0	2	46	106	214	100	48	5	0	0	(13)	
(14) Wind	WSAvg	4.0	4.7	4.5	4.5	4.7	4.2	3.6	3.0	2.8	3.4	4.1	4.5	4.3	(14)	
(15) Precipitation	PrecAvg	902	24	25	47	93	107	150	115	121	87	56	46	31	(15)	
	PrecMax	1118	69	65	105	187	218	289	294	355	241	175	160	72	(16)	
	PrecMin	601	8	2	1	16	43	63	23	27	29	10	3	5	(17)	
	PrecStd	144	14	16	25	50	42	69	56	73	55	41	39	18	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	9.7	22.4	25.4	31.1	32.2	33.6	31.3	30.3	27.2	19.5	10.1	(19)	
		MCWB	3.8	7.2	15.3	15.2	20.9	22.9	24.3	24.1	21.5	18.1	13.1	7.9	(20)	
	2%	DB	3.4	5.8	16.8	22.1	27.5	29.6	31.1	29.1	27.9	23.2	15.8	6.0	(21)	
		MCWB	1.5	3.8	12.5	13.7	18.1	21.0	24.0	22.9	20.8	16.2	11.0	4.4	(22)	
	5%	DB	1.8	3.2	13.0	19.1	24.9	27.6	29.0	27.6	26.1	20.7	12.7	3.9	(23)	
		MCWB	0.6	1.6	9.6	11.9	16.9	19.9	22.5	22.1	19.6	15.4	9.0	2.7	(24)	
	10%	DB	0.2	1.3	9.6	16.9	22.6	26.0	27.5	26.0	24.0	17.8	10.3	2.3	(25)	
		MCWB	-0.4	0.1	6.8	10.7	16.3	19.3	21.5	20.9	18.4	13.0	7.0	1.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	7.6	16.5	16.5	22.5	24.3	26.7	25.7	24.0	20.4	14.1	8.1	(27)	
		MCDB	6.0	9.7	22.1	22.3	28.2	29.6	30.8	29.4	28.9	23.7	17.8	9.2	(28)	
	2%	WB	1.8	3.9	13.6	14.9	20.2	23.0	24.9	24.1	22.2	18.1	11.8	4.8	(29)	
		MCDB	3.1	5.6	16.2	20.1	24.9	27.4	28.7	27.4	26.2	22.0	14.7	5.7	(30)	
	5%	WB	0.7	1.8	9.6	13.1	18.6	21.7	23.6	23.0	20.8	15.9	9.2	2.7	(31)	
		MCDB	1.5	3.0	13.0	17.8	23.2	25.7	27.9	26.4	24.5	19.2	12.5	3.9	(32)	
	10%	WB	-0.3	0.2	6.8	11.4	17.0	20.5	22.6	21.9	19.6	13.7	7.4	1.1	(33)	
		MCDB	0.2	1.1	9.7	16.1	20.9	24.7	26.7	25.0	23.4	17.8	9.7	2.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	7.9	8.1	9.7	10.0	9.6	9.7	9.5	10.5	9.4	7.9	6.8	(35)	
		MCDBR	8.2	8.8	12.6	14.0	12.8	11.5	11.1	10.6	12.0	13.1	11.4	8.2	(36)	
	5% WB	MCWBR	7.1	7.2	8.0	7.9	6.2	5.6	5.4	5.6	6.1	7.2	7.6	6.9	(37)	
		MCWBR	7.8	8.5	11.7	12.2	10.7	9.9	10.1	9.6	10.3	10.6	10.7	7.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.264	0.293	0.317	0.362	0.390	0.400	0.414	0.399	0.368	0.323	0.289	0.262	(40)	
		taud	2.401	2.319	2.391	2.315	2.297	2.301	2.261	2.316	2.376	2.490	2.505	2.447	(41)	
	Ebn at Noon		879	902	925	901	882	871	853	855	856	860	844	849	(42)	
		Edh at Noon	75	97	104	122	128	128	132	121	105	82	68	65	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.77	2.82	3.77	4.56	5.39	6.01	6.23	5.29	4.24	2.73	1.80	1.32	(44)	
	RadStd		0.18	0.29	0.40	0.46	0.41	0.45	0.28	0.36	0.31	0.38	0.16	0.12	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (53 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page